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SynC: Synthetic Image Caption Dataset Refinement with One-to-many Mapping for Zero-shot Image Captioning

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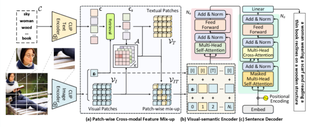
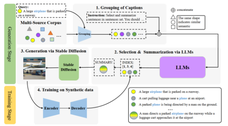
Introduction

Zero-shot Image Captioning

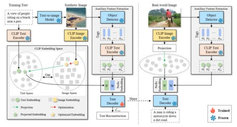
- **Zero-shot image captioning:** generating captions for images **without** any task-specific paired training data.
- Recent work [1-3] leverages **image generative models** to create training data.



[1] Fuse text feature into salient region in the synthetic image



[2] Use summarized prompt for generating multi-context image.



[3] Align train-test phase with synthetic image

Problem Definition

1. Misaligned synthetic image caption pair

- Generated images often **misalign** with their input captions.
- Training on **misaligned pairs degrades** model performance.

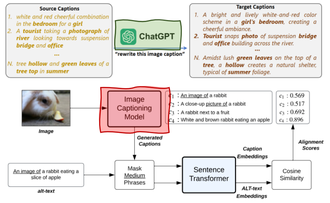
Wrong quantity

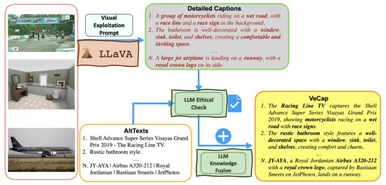
Omitting objects

Failed image

2. Limits of Web Data Pruning

- Existing methods [4–6] depend on **heavy models** (captioning models or large LLM/VLMs).
- Crucially, they assume **noisy captions**, whereas in synthetic data the **dominant noise is in images**.



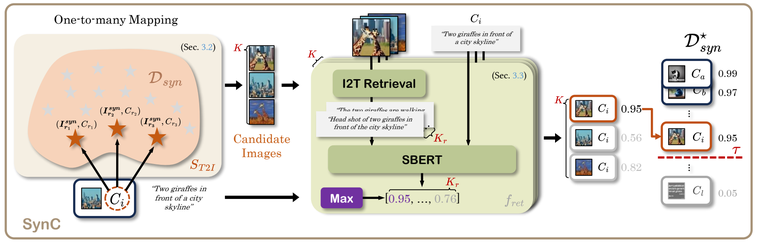


3. Goal

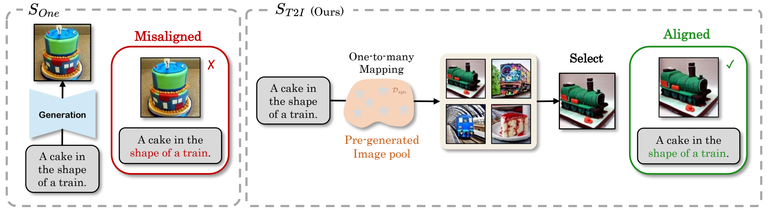
- A **lightweight**, **image-noise–focused** filter **tailored to synthetic data for ZIC**.

Proposed Framework

SynC: Without re-generating synthetic images or time-consuming pipelines, **reassign** a **better-matching image** to each caption and **filter the remaining noisy pairs**.



- **Image Selection (One-to-many Mapping)**



- **Problem:** (Naïve 1:1 mapping): misaligned pairs are simply discarded.
- **Solution:** Find **candidate images** for a given query caption. Misaligned pair can be **refined** later.

$$S_{T2I}(C_i) = \{I_r^{syn}\}_{r \in R_i},$$
$$\text{where } R_i = \underset{j \in [1,N]}{Top - K}\{\langle \mathcal{E}_I(I_j^{syn}), \mathcal{E}_T(C_i)\rangle\}.$$

I^{syn}: Synthetic Image
C: Caption
$\mathcal{E}_I, \mathcal{E}_T$: VLM encoder

- **Alignment Score Calculation**
  - Compute a caption–image alignment score using a **cycle-consistency–inspired**, **retrieval-based scorer**.

$$f_{ret}(I^{syn}, C) = \max_{r \in \hat{R}(I^{syn})} \langle \mathcal{E}_S(C_r), \mathcal{E}_S(C)\rangle,$$
$$\text{where } \hat{R}(I^{syn}) = \underset{j \in [1,N]}{Top - K_r}\{\langle \mathcal{E}_I(I^{syn}), \mathcal{E}_T(C_j)\rangle\}.$$

$\mathcal{E}_S$: Sentence BERT

- **Final Filtering**
  - **Rank** by alignment score **and keep the top τ fraction**.

$$SynC(\mathcal{D}_{syn}, \tau) = \mathcal{D}^*_{syn}.$$

Experimental Results

Evaluation on In-domain / Cross-domain Setup

Method	COCO					Flickr30k				
	B@4	M	R	C	S	B@4	M	R	C	S
ViT-B/32										
DeCap [27] ICLR'23	24.7	25.0	–	91.2	18.7	21.2	21.8	–	56.7	15.2
ViECap [13] ICCV'23	27.2	24.8	–	92.9	18.2	21.4	20.1	–	47.9	13.6
ICSD [33] AAAI'24	29.9	25.4	–	96.6	–	25.2	20.6	–	54.3	–
SynTIC [30] AAAI'24	29.9	25.8	53.2	101.1	19.3	22.3	22.4	47.3	56.6	16.6
IFCap [23] EMNLP'24	30.8	**26.7**	–	108.0	20.3	23.5	23.0	–	64.4	17.0
Baseline [32] ECCV'24	31.5	25.9	53.9	103.8	19.7	26.9	23.0	50.1	61.3	16.8
+SynC	**33.6**	**26.7**	**55.3**	**112.0**	**20.5**	**28.2**	**23.4**	**50.5**	**65.8**	**17.1**
ViT-L/14										
CgT-GAN† [53] ACMMM'23	30.3	26.9	54.5	108.1	20.5	24.1	22.6	48.2	64.9	16.1
Baseline [32] ECCV'24	33.6	26.9	55.4	113.6	20.8	28.5	24.3	51.4	69.5	18.2
+SynC	**35.2**	**27.8**	**56.5**	**119.8**	**21.9**	**29.6**	**24.8**	**52.5**	**75.6**	**18.5**

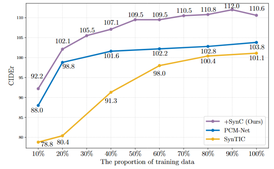
Method	COCO ⟹ Flickr30k					Flickr30k ⟹ COCO				
	B@4	M	R	C	S	B@4	M	R	C	S
ViT-B/32										
DeCap [27]	16.3	17.9	–	35.7	11.1	12.1	18.0	–	44.4	10.9
ViECap [13]	17.4	18.0	–	38.4	11.2	12.6	19.3	–	54.2	12.5
SynTIC [30]	17.9	18.6	42.7	38.4	11.9	14.6	19.4	40.9	47.0	11.9
IFCap [23]	17.8	19.4	–	47.5	12.7	14.7	**20.4**	–	**60.7**	**13.6**
Baseline [32]	20.8	19.2	45.2	45.5	12.9	17.1	19.6	43.2	54.9	12.8
+SynC	**22.4**	**20.0**	**45.9**	**49.5**	**13.4**	**18.4**	20.0	**44.1**	58.4	13.4
ViT-L/14										
CgT-GAN† [53]	17.3	19.6	43.9	47.5	12.9	15.2	19.4	40.9	58.7	13.4
Baseline [32]	23.9	21.2	47.8	55.9	14.2	17.9	20.3	44.0	61.3	13.5
+SynC	**25.2**	**22.1**	**49.6**	**60.3**	**15.8**	**19.9**	**21.3**	**45.6**	**66.8**	**14.9**

Web data filtering / computational cost

Method	COCO				
	B@4	M	R	C	S
Augment-based pruning					
VeCLIP [22] ECCV'24	27.0	25.9	52.5	93.9	19.8
LaCLIP [12] NeurIPS'23	23.7	22.6	49.6	80.5	16.3
Recaptioning [15] ArXiv'24	27.8	25.8	52.8	95.1	19.2
Metric-based pruning					
Sieve [34] CVPR'24	32.4	25.8	54.1	106.3	**19.9**
Sieve + S_{T2I}	**32.9**	**26.1**	**54.6**	**108.1**	**19.9**
CLIPScore [16] EMNLP'21	31.6	25.8	53.7	104.4	19.5
CLIPScore + S_{T2I}	**32.3**	**26.0**	**54.3**	**106.5**	**19.8**

Method	Preprocess	$\mathcal{L}(\cdot)$	S	f	Total	Δ
VeCLIP [22]	46.5h	–	–	–	46.5h	-9.9
LaCLIP [12]	28h	–	–	–	28h	-23.3
Recaptioning [15]	36h	–	–	–	36h	-8.7
Sieve [34]	36h	0.7h	–	0h	36.7h	+2.5
CLIPScore [16]	–	0.9h	–	0h	0.9h	+0.6
SynC	–	1.5h	0.4h	0.4h	2.3h	+8.2

Ablation on τ



Qualitative Result

References

[1] Luo, Jianjie, et al. "Unleashing Text-to-Image Diffusion Prior for Zero-Shot Image Captioning." ECCV '24.
[2] Ma, Feipeng, et al. "Image captioning with multi-context synthetic data." AAAI '24.
[3] Liu, Zhiyue, et al. "Improving cross-modal alignment with synthetic pairs for text-only image captioning." AAAI '24
[4] Fan, Lijie, et al. "Improving clip training with language rewrites." NeurIPS '23: 35544-35575.
[5] Mahmoud, Anas, et al. "Sieve: Multimodal dataset pruning using image captioning models." CVPR '24.
[6] Lai, Zhengfeng, et al. "Veclip: Improving clip training via visual-enriched captions." ECCV '24.